TALENTRA DATABASE DESIGN HACKATHON

Milestone 9 — Column Inventories (B30–B32)

Status: Proposed column baseline for team review. Not a complete final data dictionary; reconcile each table with the approved Milestone 7 entity list and Milestone 8 relationships.

1. Milestone scope

The docket asks for column lists across Administration, Organisation and account, and Gig marketplace. Each column should have data type/domain, nullability, and sensitive-data classification. These proposed columns are a starting point, not a claim that every field is required.

2. Shared conventions

Convention	Baseline
Identifiers	Proposed PostgreSQL BIGINT identity PK named id, consistently used; confirm team decision.
Names	snake_case columns; FK named after referenced table plus _id.
Audit	Business tables: created_at, created_by, updated_at, updated_by; use UTC timestamps. History/audit tables have documented append-only policy.
Nullable values	Every nullable column needs a reason; distinguish unknown from not applicable.
Sensitive data	Classify every column Public, Internal, Personal, or Sensitive. Do not store plaintext secrets or payment-card details.
Shared lookups	Use Master Data for fixed codes where that module owns the domain; include code, localized display name, active flag and sort order in lookup design.

3. B30 — Administration (Member A)

Entity	Proposed columns (type; nullability; classification / notes)
app_user	id BIGINT NOT NULL PK; email VARCHAR(254) NULL; phone_e164 VARCHAR(16) NULL; account_status_code VARCHAR(40) NOT NULL; created_at TIMESTAMPTZ NOT NULL; created_by BIGINT NULL; updated_at TIMESTAMPTZ NOT NULL; updated_by BIGINT NULL; deleted_at TIMESTAMPTZ NULL. Email/phone Personal; status Internal. Decide identity uniqueness and whether email/phone are required.
credential	id BIGINT NOT NULL PK; user_id BIGINT NOT NULL FK; credential_type_code VARCHAR(40) NOT NULL; password_hash TEXT NULL; provider_subject VARCHAR(255) NULL; created_at TIMESTAMPTZ NOT NULL; revoked_at TIMESTAMPTZ NULL. Sensitive: hash/provider subject. Never store plaintext password. Define mutually exclusive credential fields and unique identity.
role	id BIGINT NOT NULL PK; code VARCHAR(80) NOT NULL; display_name VARCHAR(150) NOT NULL; active_flag BOOLEAN NOT NULL DEFAULT TRUE; sort_order INTEGER NOT NULL DEFAULT 0; audit fields. Code Internal; display name Internal.
permission	id BIGINT NOT NULL PK; code VARCHAR(100) NOT NULL; display_name VARCHAR(150) NOT NULL; active_flag BOOLEAN NOT NULL DEFAULT TRUE; sort_order INTEGER NOT NULL DEFAULT 0; audit fields. Internal.
user_role	id BIGINT NOT NULL PK; user_id BIGINT NOT NULL FK; role_id BIGINT NOT NULL FK; assigned_at TIMESTAMPTZ NOT NULL; assigned_by BIGINT NULL; audit fields. Internal; pair uniqueness if duplicate assignment disallowed.
role_permission	id BIGINT NOT NULL PK; role_id BIGINT NOT NULL FK; permission_id BIGINT NOT NULL FK; audit fields. Internal; pair uniqueness if duplicates disallowed.
user_session	id BIGINT NOT NULL PK; user_id BIGINT NOT NULL FK; device_id VARCHAR(200) NULL; started_at TIMESTAMPTZ NOT NULL; last_activity_at TIMESTAMPTZ NOT NULL; ended_at TIMESTAMPTZ NULL; revoked_at TIMESTAMPTZ NULL; audit fields. Device/session identifiers Sensitive or Internal based on policy.
consent_record	id BIGINT NOT NULL PK; user_id BIGINT NOT NULL FK; purpose_code VARCHAR(80) NOT NULL; consent_version VARCHAR(40) NOT NULL; decision_code VARCHAR(30) NOT NULL; recorded_at TIMESTAMPTZ NOT NULL; withdrawn_at TIMESTAMPTZ NULL; audit fields. Personal; retain history per policy.
audit_event	id BIGINT NOT NULL PK; actor_user_id BIGINT NULL FK; action_code VARCHAR(80) NOT NULL; entity_type VARCHAR(100) NOT NULL; entity_id VARCHAR(100) NULL; occurred_at TIMESTAMPTZ NOT NULL; correlation_id VARCHAR(100) NULL; metadata JSONB NULL. Internal/Sensitive depending on metadata; append-only and redact secrets.
block_entry	id BIGINT NOT NULL PK; blocker_user_id BIGINT NOT NULL FK; blocked_user_id BIGINT NOT NULL FK; reason_code VARCHAR(60) NULL; created_at TIMESTAMPTZ NOT NULL; created_by BIGINT NULL; updated_at TIMESTAMPTZ NOT NULL; updated_by BIGINT NULL; deleted_at TIMESTAMPTZ NULL. Personal/Internal; include only if agreed.

4. B31 — Organisation and account (Member B)

Entity	Proposed columns (type; nullability; classification / notes)
organisation	id BIGINT NOT NULL PK; legal_name VARCHAR(200) NOT NULL; display_name VARCHAR(200) NULL; status_code VARCHAR(40) NOT NULL; industry_id BIGINT NULL FK to Master Data; created_at TIMESTAMPTZ NOT NULL; created_by BIGINT NULL; updated_at TIMESTAMPTZ NOT NULL; updated_by BIGINT NULL; deleted_at TIMESTAMPTZ NULL. Names Internal; legal data Internal.
organisation_location	id BIGINT NOT NULL PK; organisation_id BIGINT NOT NULL FK; location_name VARCHAR(150) NULL; address_line_1 VARCHAR(200) NOT NULL; address_line_2 VARCHAR(200) NULL; city_id BIGINT NULL FK to Master Data; state_id BIGINT NULL FK; country_id BIGINT NOT NULL FK; postal_code VARCHAR(20) NULL; created_at TIMESTAMPTZ NOT NULL; created_by BIGINT NULL; updated_at TIMESTAMPTZ NOT NULL; updated_by BIGINT NULL; deleted_at TIMESTAMPTZ NULL. Personal/Internal depending on address use.
organisation_identifier	id BIGINT NOT NULL PK; organisation_id BIGINT NOT NULL FK; country_id BIGINT NOT NULL FK; identifier_type_code VARCHAR(40) NOT NULL; identifier_value VARCHAR(200) NOT NULL; created_at TIMESTAMPTZ NOT NULL; created_by BIGINT NULL; updated_at TIMESTAMPTZ NOT NULL; updated_by BIGINT NULL; deleted_at TIMESTAMPTZ NULL. Sensitive; protect/mask; country/type required.
organisation_verification	id BIGINT NOT NULL PK; organisation_id BIGINT NOT NULL FK; status_code VARCHAR(40) NOT NULL; evidence_reference VARCHAR(500) NULL; verified_at TIMESTAMPTZ NULL; verified_by BIGINT NULL; reason_code VARCHAR(80) NULL; audit fields. Sensitive/Internal; avoid embedding secrets in evidence.
recruiter_organisation_mapping	id BIGINT NOT NULL PK; organisation_id BIGINT NOT NULL FK; recruiter_user_id BIGINT NOT NULL FK to Administration; valid_from TIMESTAMPTZ NOT NULL; valid_to TIMESTAMPTZ NULL; created_at TIMESTAMPTZ NOT NULL; created_by BIGINT NULL; updated_at TIMESTAMPTZ NOT NULL; updated_by BIGINT NULL. Internal; timezone UTC; no overlapping active mapping unless allowed.
account	id BIGINT NOT NULL PK; account_code VARCHAR(80) NOT NULL; organisation_id BIGINT NULL FK; status_code VARCHAR(40) NOT NULL; created_at TIMESTAMPTZ NOT NULL; created_by BIGINT NULL; updated_at TIMESTAMPTZ NOT NULL; updated_by BIGINT NULL; deleted_at TIMESTAMPTZ NULL. Internal; confirm whether account is tenant/billing/login boundary.
account_admin	id BIGINT NOT NULL PK; account_id BIGINT NOT NULL FK; user_id BIGINT NOT NULL FK to Administration; valid_from TIMESTAMPTZ NOT NULL; valid_to TIMESTAMPTZ NULL; created_at TIMESTAMPTZ NOT NULL; created_by BIGINT NULL; updated_at TIMESTAMPTZ NOT NULL; updated_by BIGINT NULL. Internal; uniqueness and overlapping rules need decision.
organisation_hierarchy	id BIGINT NOT NULL PK; parent_organisation_id BIGINT NOT NULL FK; child_organisation_id BIGINT NOT NULL FK; valid_from DATE NULL; valid_to DATE NULL; audit fields. Internal; prevent self-parenting and cycles if hierarchy is required.

5. B32 — Gig marketplace (Member C)

Entity	Proposed columns (type; nullability; classification / notes)
gig_worker_profile	id BIGINT NOT NULL PK; user_id BIGINT NOT NULL FK to Administration; headline VARCHAR(200) NULL; bio TEXT NULL; status_code VARCHAR(40) NOT NULL; country_id BIGINT NULL FK to Master Data; city_id BIGINT NULL FK; audit fields; deleted_at TIMESTAMPTZ NULL. Personal/Internal; bio needs a justified max length or documented unbounded content.
gig_service	id BIGINT NOT NULL PK; gig_worker_profile_id BIGINT NOT NULL FK; service_code VARCHAR(80) NULL; title VARCHAR(150) NOT NULL; description TEXT NULL; active_flag BOOLEAN NOT NULL DEFAULT TRUE; audit fields; deleted_at TIMESTAMPTZ NULL. Internal/Public based on marketplace visibility.
gig_availability	id BIGINT NOT NULL PK; gig_worker_profile_id BIGINT NOT NULL FK; starts_at TIMESTAMPTZ NOT NULL; ends_at TIMESTAMPTZ NOT NULL; timezone_name VARCHAR(80) NOT NULL; availability_status_code VARCHAR(40) NOT NULL; audit fields. Personal/Internal; end must be after start; clarify recurrence support.
gig_request	id BIGINT NOT NULL PK; requester_user_id BIGINT NULL FK; organisation_id BIGINT NULL FK; title VARCHAR(200) NOT NULL; description TEXT NOT NULL; starts_at TIMESTAMPTZ NOT NULL; ends_at TIMESTAMPTZ NOT NULL; pricing_model_code VARCHAR(40) NOT NULL; price_amount NUMERIC(12,2) NULL; currency_code CHAR(3) NULL; status_code VARCHAR(40) NOT NULL; audit fields; deleted_at TIMESTAMPTZ NULL. Require currency when price supplied; decide requester user/org rule.
gig_application	id BIGINT NOT NULL PK; gig_request_id BIGINT NOT NULL FK; gig_worker_profile_id BIGINT NOT NULL FK; status_code VARCHAR(40) NOT NULL; applied_at TIMESTAMPTZ NOT NULL; message TEXT NULL; audit fields. Personal/Internal; decide duplicate application rule.
gig_assignment	id BIGINT NOT NULL PK; gig_application_id BIGINT NOT NULL FK; starts_at TIMESTAMPTZ NOT NULL; ends_at TIMESTAMPTZ NOT NULL; status_code VARCHAR(40) NOT NULL; agreed_amount NUMERIC(12,2) NULL; currency_code CHAR(3) NULL; audit fields. Internal; check end after start and amount nonnegative; require currency with amount.
gig_handoff	id BIGINT NOT NULL PK; gig_assignment_id BIGINT NOT NULL FK; external_system_code VARCHAR(80) NOT NULL; external_reference VARCHAR(200) NOT NULL; handoff_status_code VARCHAR(40) NOT NULL; occurred_at TIMESTAMPTZ NOT NULL; metadata JSONB NULL; audit fields. Internal/Sensitive; metadata must not contain secrets; define idempotency.
gig_earnings_snapshot / external reference	id BIGINT NOT NULL PK if local entity approved; gig_assignment_id BIGINT NOT NULL FK; amount NUMERIC(12,2) NOT NULL; currency_code CHAR(3) NOT NULL; source_system_code VARCHAR(80) NOT NULL; source_reference VARCHAR(200) NULL; captured_at TIMESTAMPTZ NOT NULL. Financial data Sensitive; confirm local snapshot vs external ownership.
gig_worker_list_entry	id BIGINT NOT NULL PK; owner_user_id BIGINT NOT NULL FK; gig_worker_profile_id BIGINT NOT NULL FK; created_at TIMESTAMPTZ NOT NULL; created_by BIGINT NULL; updated_at TIMESTAMPTZ NOT NULL; updated_by BIGINT NULL; deleted_at TIMESTAMPTZ NULL. Internal; include only if saved lists are in scope.
gig_rating	id BIGINT NOT NULL PK; gig_assignment_id BIGINT NOT NULL FK; reviewer_user_id BIGINT NOT NULL FK; rated_user_id BIGINT NOT NULL FK; score SMALLINT NOT NULL; review_text VARCHAR(2000) NULL; created_at TIMESTAMPTZ NOT NULL; created_by BIGINT NULL; updated_at TIMESTAMPTZ NOT NULL; updated_by BIGINT NULL; deleted_at TIMESTAMPTZ NULL. Personal/Public visibility needs decision; score range and one/multiple ratings must be defined.

6. Milestone 9 validation checklist

• Every entity matches the approved Milestone 7 list and each FK matches the approved Milestone 8 relationship model.

• Every column has a type, nullability, purpose, and Public/Internal/Personal/Sensitive classification in the complete data dictionary.

• Every nullable field has a written justification; timestamps use UTC; dates and time ranges use consistent domains.

• Money is fixed-precision NUMERIC with currency code; sensitive identifiers and credential data have explicit protection rules.

• Business tables include required audit fields; history and audit tables have append-only/retention rules.

• Open decisions are recorded instead of silently assumed; team owners accept/revise/remove each proposed entity and column.

• Milestone 9 output is three reviewed column lists; keys and constraints are finalized in Milestone 10.

Important: The docket requires column lists with types/domains, nullability and sensitive-data classification. Detailed fields above are proposals; do not treat them as mandated or approved until the module owners confirm them.